

ODESA, Ukraine

WMO: 338370

Lat: 46.4408N

Lon: 30.7703E

Elev: 138

StdP: 14.62

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	13.5	1.8	6.1	9.1	6.9	7.9	14.6	29.6	24.2	26.0	26.4	10.4	20	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	91.2	69.2	87.9	68.3	84.8	67.7	73.6	82.3	72.0	81.1	70.6	79.8	8.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	115.6	77.6	69.4	108.6	76.4	67.7	102.4	75.3	37.3	82.2	35.8	80.9	34.6	79.6	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.0	17.9	DB	1.3	96.2	4.7	3.9	-2.1	99.0	-4.8	101.2	-7.5	103.4	-10.9	106.2	(4)
(5)			WB	0.7	77.2	4.7	2.0	-2.7	78.6	-5.4	79.8	-8.0	80.9	-11.4	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	30.0	32.5	39.8	50.6	61.5	69.7	74.4	73.7	63.5	52.3	43.2	33.9	(6)	
	DBStd	17.19	8.96	8.86	7.16	6.24	6.21	5.61	5.00	5.79	6.33	7.37	8.20	8.93	(7)	
	HDD50	2288	619	491	320	63	2	0	0	0	1	60	231	501	(8)	
	HDD65	5490	1084	911	781	432	144	19	1	4	100	396	654	964	(9)	
	CDD50	3096	0	0	5	82	359	590	757	736	406	132	28	1	(10)	
	CDD65	823	0	0	0	1	36	159	294	275	55	3	0	0	(11)	
	CDH74	6727	0	0	0	17	246	1201	2564	2392	301	6	0	0	(12)	
	CDH80	2181	0	0	0	3	44	329	878	873	54	0	0	0	(13)	
	WSAvg	8.4	9.2	9.4	9.6	8.6	7.6	7.7	7.5	7.4	7.8	8.3	9.0	9.0	(14)	
	PrecAvg	18.1	1.6	1.3	1.3	1.1	1.5	2.0	1.8	1.6	1.6	1.5	1.5	1.5	(15)	
(16) Precipitation	PrecMax	28.3	4.4	3.5	3.1	2.6	3.9	4.2	4.8	5.4	4.6	6.8	4.2	4.7	(16)	
	PrecMin	13.4	0.3	0.0	0.0	0.0	0.1	0.5	0.1	0.1	0.0	0.2	0.0	0.0	(17)	
	PrecStd	3.9	1.1	0.9	0.9	0.7	1.0	0.9	1.3	1.4	1.4	1.3	1.0	1.1	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	53.9	64.3	75.7	84.6	90.0	95.1	96.7	85.7	73.8	64.4	55.2	(19)	
		MCWB	44.8	47.1	51.8	60.3	64.8	68.2	68.9	70.1	65.2	62.1	56.8	51.1	(20)	
	2%	DB	46.5	48.6	57.3	68.4	80.2	86.7	91.1	91.6	80.7	70.0	60.1	51.8	(21)	
		MCWB	43.9	44.1	47.7	55.4	64.4	68.0	69.6	68.9	63.9	62.0	56.7	49.1	(22)	
	5%	DB	44.0	45.8	53.3	64.5	76.6	84.0	87.8	88.1	77.2	66.4	57.3	48.9	(23)	
		MCWB	42.2	42.7	45.7	53.3	62.9	67.2	68.9	68.2	63.8	60.3	54.3	47.2	(24)	
	10%	DB	41.2	43.0	49.8	60.8	73.1	80.7	84.6	84.5	74.0	63.6	55.1	46.5	(25)	
		MCWB	39.8	40.9	43.8	51.7	61.4	66.4	68.3	67.7	63.3	58.3	52.6	45.0	(26)	
	0.4%	WB	46.8	48.4	52.4	61.2	69.1	74.1	75.6	76.4	71.8	66.5	59.8	52.5	(27)	
		MCDB	48.3	53.4	61.7	72.4	77.0	83.1	82.6	84.9	76.6	70.1	62.2	54.2	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.1	45.0	49.1	57.3	66.7	71.8	73.7	73.5	69.2	63.9	57.2	50.0	(29)	
		MCDB	45.6	47.3	56.2	65.5	75.2	80.3	82.2	82.5	74.3	67.7	59.7	51.0	(30)	
	5%	WB	42.1	43.0	46.5	55.0	64.9	70.1	72.2	71.8	67.1	61.7	55.0	47.3	(31)	
		MCDB	43.4	44.7	51.0	61.7	73.3	78.3	81.5	81.3	73.3	65.4	57.1	48.7	(32)	
	10%	WB	39.8	40.9	44.6	52.8	63.1	68.6	70.7	70.1	65.1	59.1	52.9	44.9	(33)	
		MCDB	41.1	42.7	47.9	59.3	70.8	77.0	80.5	80.0	71.4	62.6	54.9	46.3	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.5	13.8	15.8	17.6	18.0	18.8	18.6	17.0	14.5	11.3	10.7	(35)	
		MCDBR	12.5	15.3	20.0	21.1	22.3	21.5	21.6	22.0	20.6	18.0	13.5	12.6	(36)	
		MCWBR	10.1	11.2	12.4	11.3	10.7	9.6	8.7	9.0	10.8	11.3	9.8	10.3	(37)	
	5% WB	MCDBR	11.5	14.1	17.3	18.9	20.0	18.5	18.8	18.8	17.5	15.4	11.4	11.4	(38)	
		MCWBR	9.9	11.0	11.7	11.1	10.5	9.8	9.2	9.6	11.1	11.0	8.9	9.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.343	0.370	0.398	0.443	0.419	0.417	0.444	0.444	0.413	0.382	0.361	0.341	(40)	
	taud		2.314	2.260	2.227	2.127	2.288	2.326	2.261	2.248	2.298	2.358	2.347	2.340	(41)	
	Ebn at Noon		231	249	259	258	270	270	260	255	252	242	223	218	(42)	
	Edh at Noon		25	32	38	46	41	39	42	40	35	29	24	22	(43)	
(44) All-Sky Solar Radiation	RadAvg		323	612	1036	1575	2012	2123	2115	1883	1330	822	390	276	(44)	
	RadStd		57	82	98	131	153	127	115	99	145	92	59	52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.24	N/A	N/A	+1.85	+0.85	+0.76	-207	-275	+214	+149
(47)	Regional (9 neighbors)	+1.13	N/A	N/A	+1.42	N/A	N/A	-182	-254	+196	+137

Nomenclature: See separate page